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IEEE Author Profile

Jie Zhou

Ph.D. in Information and Communication Engineering

Education

Dept. of Electronic Engineering, Harbin Institute of Technology Harbin, China
Ph.D. in Information and Communication Engineering Sept. 2021 – Sept. 2026

- Integrated Master–Ph.D. Program
- Supervisor: Prof. Junhao Xie
- Dissertation: *Large-dimensional Statistical Signal Processing for Radar Target Detection*

Dept. of Electronic Engineering, Harbin Institute of Technology Harbin, China
B.E. in Electronic Information Engineering Sept. 2017 – Jun. 2021

Research Summary

My research focuses on statistical radar signal processing, particularly adaptive detection, covariance and precision matrix estimation, and performance analysis using random matrix theory. I have developed and analysed CFAR detection methods under limited training data, Gaussian and compound-Gaussian clutter, and steering-vector mismatch. My work combines theoretical analysis, Monte Carlo simulations, and validation using measured radar data. I am interested in extending these methods to multistatic radar detection and automotive sensing.

Publication record: 17 publications, including 9 first-author papers, of which 5 appeared in IEEE Transactions journals.

Publications (First Author)

1. **Jie Zhou**, Junhao Xie, and Jiaqi Chen. Direct estimation of large dimensional precision matrix. **IEEE Transactions on Signal Processing**, vol. 74, pp. 183–199, 2026.
2. **Jie Zhou**, Junhao Xie. On the limiting distribution of the likelihood ratio statistic in expected likelihood AMF detector under large-dimensional regime. **IEEE Transactions on Aerospace and Electronic Systems**, vol. 61, no. 5, pp. 15130–15137, 2025.
3. **Jie Zhou**, Junhao Xie. Performance analysis of linearly combined order statistics CFAR processors in heterogeneous background. **IEEE Transactions on Aerospace and Electronic Systems**, vol. 60, no. 2, pp. 2428–2437, 2024.
4. **Jie Zhou**, Junhao Xie. Robust CFAR detector based on KLQ estimator for multiple-target scenario. **IEEE Transactions on Geoscience and Remote Sensing**, vol. 61, pp. 1–16, 2023.
5. **Jie Zhou**, Junhao Xie, et al. Robust sliding window CFAR detection based on quantile truncated statistics. **IEEE Transactions on Geoscience and Remote Sensing**, vol. 60, pp. 1–23, 2022.
6. **Jie Zhou**, Junhao Xie. High-dimensional asymptotic analysis of adaptive radar detectors under Gaussian and compound-Gaussian clutter. **Signal Processing**, vol. 249, Paper No. 110771, 2026.
7. **Jie Zhou**, Junhao Xie. An improved quantile estimator with its application in CFAR detection. **IEEE Geoscience and Remote Sensing Letters**, vol. 20, pp. 1–5, 2023.
8. **Jie Zhou**, Junhao Xie, and Jiaqi Chen. Asymptotic performance of low rank adaptive normalized matched filter test under large dimensional regime. *2025 IEEE International Radar Conference*, Paper No. 1571099367, May 5–9, 2025, Atlanta, Georgia, USA. **Student Paper Award Finalist**.
9. **Jie Zhou**, Junhao Xie. Robust CFAR diagonally loaded AMF for mismatched signals. Accepted by *2026 CIE International Conference on Radar*, 2026.

Publications (Co-author)

10. Xingxing Liao, Junhao Xie, and **Jie Zhou**. Compound Gaussian radar clutter model with positive tempered alpha-stable texture. **IEEE Transactions on Aerospace and Electronic Systems**, vol. 61, no. 6, pp. 17364–17376, 2025.
11. Xingxing Liao, Junhao Xie, and **Jie Zhou**. A flexible-tailed model for radar sea clutter amplitudes based on the smoothly truncated Lévy flight. **IEEE Transactions on Aerospace and Electronic Systems**, vol. 61, no. 4, pp. 10258–10272, 2025.
12. Xingxing Liao, Junhao Xie, and **Jie Zhou**. A data-driven optimization method for simulating arbitrarily distributed and spatial-temporal correlated radar sea clutter. **IEEE Transactions on Geoscience and Remote Sensing**, vol. 61, pp. 1–15, 2023.
13. Xingxing Liao, Junhao Xie, and **Jie Zhou**. Compound-Gaussian spatial-temporal correlated complex clutter simulation based on a two-step data-driven method. **IEEE Transactions on Aerospace and Electronic Systems**, vol. 59, no. 6, pp. 9512–9526, 2023.
14. Xingxing Liao, **Jie Zhou**, and Junhao Xie. Generation of random sea clutter amplitude sequence with quantitative control on distribution tail. *Electronics Letters*, vol. 59, no. 20, pp. 1–3, 2023.
15. Baiqiang Zhang, **Jie Zhou**, Junhao Xie, et al. Weighted likelihood CFAR detection for Weibull background. *Digital Signal Processing*, vol. 115, Paper No. 103079, 2021.
16. Baiqiang Zhang, Junhao Xie, and **Jie Zhou**. Slow-time randomly missing data reconstruction for skywave over-the-horizon radar. *2021 CIE International Conference on Radar*, pp. 3255–3260, 2021.
17. C. A. Schone et al. (including **Jie Zhou**), 2025 IEEE International Radar Conference Student Paper Competition. IEEE Aerospace and Electronic Systems Magazine, vol. 41, no. 1, pp. 56–68, 2026.

Research Experience

Measured Radar Data Processing and Target Detection

- Developed an end-to-end multichannel radar signal processing pipeline, including beamforming, Doppler processing, adaptive detection, CFAR processing, clustering, and target tracking using measured radar data.
- Implemented and evaluated sensing algorithms under practical multichannel radar environments, with emphasis on robust detection, covariance estimation, and performance evaluation under clutter and interference.

Short Academic Visits

Aalto University, Helsinki, Finland
Academic Exchange *Aug. 2024*

Katholieke Universiteit Leuven, Leuven, Belgium
Academic Exchange *Aug. 2024*

Awards and Honors

- **National Scholarship**, Ministry of Education of China *2025*
- **2025 IEEE International Radar Conference Student Paper Award Finalist**, Atlanta, GA, USA *2025*
- **Excellent Undergraduate Thesis Award**, Harbin Institute of Technology *2021*
- **Outstanding Graduate Award**, Harbin Institute of Technology *2021*

Technical Skills

Signal Processing: Radar signal processing, adaptive detection, CFAR detection, array signal processing, beamforming.

Mathematical Tools: Random Matrix Theory (RMT), high-dimensional statistics, statistical inference, covariance estimation, detection and estimation theory.

Programming: `MATLAB`, `Python`.

Academic Services

Reviewer for: *IEEE Transactions on Aerospace and Electronic Systems*,
IEEE Transactions on Radar Systems,
IEEE Geoscience and Remote Sensing Letters,
IEEE Antennas and Wireless Propagation Letters,
IEEE Signal Processing Letters.